

APEX: Latent-Space Kernel Fusion for Selective Prediction with Frozen Medical Image Classifiers

Mohammad Kamal Abdulaziz

Dept. of Data Science and Artificial Intelligence
King Abdullah II School of Information Technology
The University of Jordan, Amman, Jordan
moh0217526@ju.edu.jo

Rizik Al-Sayyed

Department of Information Technology
King Abdullah II School of Information Technology
The University of Jordan, Amman, Jordan
r.alsayyed@ju.edu.jo

Abstract—A diagnostic model allowed to abstain is judged less by how often it is right than by how well its confidence orders its own mistakes. We present APEX, a post-hoc layer that improves that ordering without touching the network. APEX leaves the backbone frozen and works on the penultimate representation the classifier computes and then discards: it concatenates that latent vector with image descriptors, projects the pair onto a shared low-dimensional subspace, evaluates three kernel experts there, and mixes their posteriors with the original softmax under a confidence-conditional gate. The weights are fitted by minimising negative log-likelihood, a strictly proper rule, and we show that choice is what keeps the fused probabilities honest. Across six imaging modalities, three backbone families and five-fold cross-validation (90 trained heads, 18 paired configurations, fifteen metrics), APEX lowers the area under the risk-coverage curve by 44.3% and selective risk at 90% coverage from 10.71% to 6.52%. AURC, Brier score and log-likelihood improve in all 18 configurations against each of six baselines at $p < 10^{-4}$. Expected calibration error does not, and that asymmetry is the point: selective risk is invariant to monotone rescaling of confidence where calibration error is not, and in 15 of 18 configurations the lowest-calibration-error method is not the lowest-selective-risk one.

Index Terms—selective prediction, confidence calibration, post-hoc methods, medical image analysis, kernel methods, proper scoring rules

I. INTRODUCTION

A deployed diagnostic model is rarely asked to decide everything. The realistic workflow is triage: the system handles the cases it is sure about and refers the rest to a clinician [1]. The decisive question is then not aggregate accuracy but whether the model’s confidence separates the cases it will get right from the ones it will get wrong—a property called selective prediction, summarised by the area under the risk-coverage curve [2], [3].

Post-hoc calibration is the natural place to look for it: a calibrator adjusts a trained model’s probabilities on a small held-out split, costs minutes rather than GPU-days, and leaves the network untouched—which matters when the network is a frozen foundation backbone or a certified artefact that cannot be retrained [4], [5]. The literature that developed these methods, however, is organised around a different target. Expected calibration error (ECE) asks whether confidence matches accuracy *on average within confidence bands*; selective risk asks whether confidence *orders* the errors. The two

coincide in the ideal case and come apart in practice, because a monotone transform of the confidence score leaves every risk-coverage curve unchanged while moving ECE (Proposition 1).

This paper asks what is available to a post-hoc method if it optimises for ordering rather than for average agreement, and the answer starts from what the frozen model already computes. A linear head reduces a d -dimensional penultimate representation to a handful of logits, discarding everything about the geometry of that representation—which regions are dense, which sit near a decision boundary, which are isolated. A layer that reads the latent vector directly can recover part of that geometry and say something the softmax cannot. We build such a layer, APEX, from three kernel experts of deliberately different inductive bias fitted on a low-dimensional projection of the latent space, and mix their posteriors with the original softmax. The construction is what is new: the components are individually standard, but stacking them over a shared subspace and gating the base model on its own confidence yields a score that ranks errors far better than any of its parts. Our contributions are four. We give a **post-hoc fusion layer for frozen backbones** that reduces AURC by 44.3% and wins all 18 dataset $\times$ backbone configurations against each of six baselines, at the smallest p -value the Wilcoxon test can return at $n = 18$. We establish a **separation between two reliability criteria**, proving selective risk invariant under monotone reparametrisation where ECE is not, and measuring the gap: in 15 of 18 configurations the lowest-ECE method is not the lowest-AURC one. We report our own ECE plainly, including where a baseline beats it. We show **the objective governs calibration**—refitting the same experts under a focal objective instead of the strictly proper one costs 1.8×10^{-2} ECE and buys nothing. And we work from **an audited corpus**: we hash every image before use and report what we removed, 1,337 files including 798 segmentation masks a file glob had been loading as ultrasound scans.

II. BACKGROUND

A. Post-hoc calibration

Temperature scaling divides the logits by one learned scalar and is the standard reference point, being both effective and unable to change any prediction [4]. Vector and matrix scaling generalise it to per-class affine maps, isotonic regression fits a

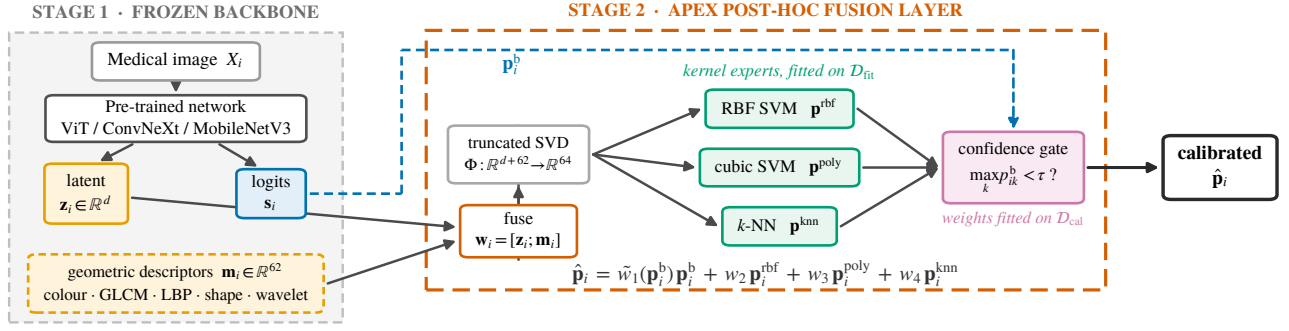


Fig. 1. APEX in full. The backbone and its linear head are frozen; everything to the right of the dashed boundary is fitted post-hoc on held-out splits and adds no gradient step to the network. Latent vector and descriptor block are projected to a shared 64-dimensional subspace by a truncated SVD, three kernel experts are evaluated on it, and their posteriors are mixed with the base softmax under a gate that discounts the base model exactly when the base model is unsure.

monotone transform per class [6], and Dirichlet calibration places a regularised log-linear model on the simplex [5]. Across the family the map acts on the model’s own output and is fitted to a criterion evaluated on that output alone.

B. Measuring calibration

ECE bins predictions by confidence and averages the gap between confidence and accuracy [7]. Binning makes the estimate optimistic—errors of opposite sign cancel inside a bin—which motivates debiased estimators [8] and class-wise variants [9]. Strictly proper scoring rules, Brier and log-loss, avoid binning and are minimised only by the true conditional distribution [10], [11]. We report all of these and treat the proper scores as primary evidence on probability quality.

C. Selective prediction and stacking

Selective classification pairs a predictor with a rejection rule [2]. Sweeping the threshold traces risk against coverage; the area under that curve (AURC) summarises how well the confidence score ranks errors [3]; its excess from subtracts the value an oracle ranking would attain. Learned reject options such as SelectiveNet [12] require training, which a frozen backbone rules out; maximum softmax probability remains the standard training-free baseline [13]. APEX is a stacked generalisation [14] whose base learners see a projection of the frozen latent space rather than the raw image, and whose meta-model is a gated convex combination fitted under a strictly proper rule.

III. METHOD

A. Setting and notation

Fig. 1 shows the construction end to end; this section defines it. Let f be a trained classifier over K classes with fixed parameters. For an image X_i we read the penultimate representation $\mathbf{z}_i \in \mathbb{R}^d$ and the softmax output $\mathbf{p}_i^b \in \Delta^{K-1}$, and compute a 62-dimensional descriptor vector $\mathbf{m}_i$ (intensity statistics, grey-level co-occurrence features, local binary patterns, shape moments, wavelet energies). Labelled data is split three ways: $\mathcal{D}_{\text{fit}}$ for the experts, $\mathcal{D}_{\text{cal}}$ for the mixture weights, and an evaluation fold no component ever sees.

B. A shared subspace for heterogeneous features

Latent coordinates and descriptors live on incompatible scales, so we standardise the concatenation $\mathbf{w}_i = [\mathbf{z}_i; \mathbf{m}_i]$ and project it to a common subspace,

$$\mathbf{u}_i = \Phi(\mathbf{w}_i), \quad \Phi: \mathbb{R}^{d+62} \rightarrow \mathbb{R}^r, \quad (1)$$

where Φ standardises each coordinate and then applies a truncated singular value decomposition to $r = 64$ components, fitted on $\mathcal{D}_{\text{fit}}$ alone. The projection is linear; every nonlinearity in APEX lives in the experts. Compression is not the point: kernel machines degrade in high dimension, and $r \ll d$ is what makes the RBF bandwidth meaningful.

C. Three experts with different inductive biases

On $\mathbf{u}$ we fit three probabilistic classifiers chosen so their errors are unlikely to coincide. An RBF support vector machine responds to local density with a smooth, globally regularised boundary; a cubic-polynomial SVM captures low-order curvature the RBF kernel smooths away; a k -nearest-neighbour classifier is purely local, and asks what a linear head cannot—are this sample’s neighbours of one class or of several? Both SVMs use $C = 1$, with the scaled RBF bandwidth and degree three respectively, and turn margins into posteriors by Platt scaling [15]; the k -NN posterior is the inverse-distance-weighted vote of the $k = 5$ nearest neighbours. We write them $\mathbf{p}_i^{\text{rbf}}$, $\mathbf{p}_i^{\text{poly}}$, $\mathbf{p}_i^{\text{knn}}$.

D. Fusion and the confidence gate

The fused posterior is a convex combination of the base softmax and the three experts,

$$\hat{\mathbf{p}}_i^\circ = w_1 \mathbf{p}_i^b + w_2 \mathbf{p}_i^{\text{rbf}} + w_3 \mathbf{p}_i^{\text{poly}} + w_4 \mathbf{p}_i^{\text{knn}}, \quad (2)$$

with $w_j \geq 0$, $\sum_j w_j = 1$. Fixed weights trust the base model equally on the cases it understands and the cases it does not, so we make the first weight a function of its own confidence. Writing $c_i = \max_k p_{ik}^b$ and fixing a threshold τ ,

$$\tilde{w}_1(c_i) = \begin{cases} w_1, & c_i \geq \tau, \\ \lambda w_1, & c_i < \tau, \end{cases} \quad 0 < \lambda < 1, \quad (3)$$

and the mass $(1 - \lambda)w_1$ freed below the threshold is re-distributed uniformly across the experts. The gated predictor $\hat{p}_i$ is Eq. 2 with w_1 replaced by $\tilde{w}_1(c_i)$, and stays on the simplex. The gate is crude on purpose: it does not work out which expert is right, it withdraws trust from the base model where the base model has already said it has none. We use $\tau = 0.85$, $\lambda = 0.20$ throughout; Section V-C shows the result is insensitive to τ .

E. Fitting the weights

The weights are obtained on $\mathcal{D}_{\text{cal}}$ by minimising negative log-likelihood,

$$\mathbf{w}^* = \arg \min_{\mathbf{w} \in \Delta^3} -\frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_{i \in \mathcal{D}_{\text{cal}}} \log \hat{p}_{i,y_i}^\circ, \quad (4)$$

a convex problem in $\mathbf{w}$. The objective is the *ungated* mixture: keeping the threshold out of it preserves convexity, and the gate, which only redistributes a fixed fraction of w_1 , is applied at inference with the weights so obtained. Negative log-likelihood is a strictly proper scoring rule, so its minimiser over a well-specified family is the true conditional distribution [11]. Objectives built to sharpen decisions rather than to score probabilities—focal loss being the common example [16]—carry no such guarantee, and Section V-C measures what substituting one costs. APEX adds no gradient step to the backbone: fitting takes under a minute per configuration on CPU for our largest dataset, and inference adds three kernel evaluations and a weighted sum.

IV. EXPERIMENTAL PROTOCOL

A. Data

Six public corpora, one per modality, 32,251 images in total after the audit below: blood smear microscopy (12,513 images, 4 classes) [17], brain MRI (7,013, 4) [18], chest radiography (3,994, 2) [19], endoscopy (3,990, 4) [20], retinal OCT (3,963, 4) [19] and breast ultrasound (778, 3) [21]. Every image carries a matched descriptor row on all six.

B. Data hygiene

Duplicate and mislabelled files inflate cross-validated results silently: copies of one image land in different folds and the model is tested on material it memorised [22]. We hashed all 33,588 candidate files and quarantined 1,337 (4.0%): 798 binary segmentation masks a file-extension glob had been loading as ultrasound scans, 531 byte-identical copies within a class, and both members of the four groups in which one image appeared under two labels. The masks were the most consequential: they halved the nominal ultrasound corpus and made its classes separable by silhouette alone. Every number here is computed after that removal.

C. Backbones and heads

We use three architecture families of different design: ViT-S/16 ($d = 384$) [23], ConvNeXt-T ($d = 768$) [24] and MobileNetV3-L ($d = 1280$) [25]. Every backbone parameter is held fixed at its ImageNet value and only a linear head with

dropout is trained, ten epochs per fold, the checkpoint selected on an inner split so the evaluation fold is never used for model selection. One caveat, stated precisely: MobileNetV3 uses batch normalisation, so its running statistics adapt to the training split—never to an evaluation fold—while its weights stay frozen. ViT and ConvNeXt normalise per layer and are unaffected.

D. Protocol and metrics

Every dataset $\times$ backbone pair is evaluated under stratified five-fold cross-validation, giving 90 trained heads and 18 paired configurations; within each fold the training portion splits 80/20 into $\mathcal{D}_{\text{fit}}$ and $\mathcal{D}_{\text{cal}}$; all fifteen metrics appear in Table II. Isotonic regression is a step function and ties 71% of its confidences, so we report AURC as the expectation over tie-breaking, which is independent of input order and coincides with the plain definition when confidences are distinct. Baselines are the uncalibrated softmax, MC dropout with 30 stochastic passes [26], temperature and vector scaling [4], isotonic regression [6] and Dirichlet calibration [5]. Following standard practice for multi-dataset comparison [27], configurations are the unit of analysis and methods are compared with the two-sided Wilcoxon signed-rank test [28]. MC dropout’s stochasticity was verified rather than assumed: the passes differed in all 90 folds.

V. RESULTS

A. Selective risk

Averaged over the 18 configurations (Table I), APEX reduces AURC from 4.86×10^{-2} to 2.71×10^{-2} —44.3% against the uncalibrated softmax, 39.0% against the strongest baseline—and risk at 90% coverage from 10.71% to 6.52%. The margin is not carried by a subset of easy cases. Fig. 2(a) shows the mean risk–coverage curve below every baseline at every coverage, the gap growing from 0.7 points of risk at 30% coverage to 3.5 points at 90%—the range a triage system would use. Fig. 2(b) makes the comparison as strict as it can be made—APEX against whichever baseline was best *for that configuration*—and all eighteen points fall below the diagonal. Fig. 2(c) shows the effect in every modality, from 18% on endoscopy to 82% on blood smears.

The proper scores move with it: Brier falls 30.6% and NLL 30.2%, each 18/18 against all six baselines at $p = 7.6 \times 10^{-6}$. Balanced accuracy rises 4.18 points, 18/18 against five of the six and 17/18 against Dirichlet ($p = 2.3 \times 10^{-5}$). Because APEX may change the arg-max this is a genuine accuracy gain rather than a rescaling—obtained with the backbone frozen.

B. Calibration is a different axis

ECE does not follow. APEX cuts it from 4.95×10^{-2} to 3.39×10^{-2} , but the calibrators land in the same place—3.28, 3.11 and 3.47×10^{-2} for temperature, vector and isotonic scaling—and the paired tests separate none of them from APEX ($p = 0.37, 0.13, 0.87$). Dirichlet calibration reaches 2.99×10^{-2} and beats APEX in 12 of the 18 configurations ($p = 0.030$). APEX also has the highest maximum calibration

TABLE I

MEAN $\pm$ STANDARD DEVIATION OVER THE 18 DATASET $\times$ BACKBONE CONFIGURATIONS. THE LAST TWO ROWS COUNT THE CONFIGURATIONS APEX WINS AND GIVE THE WORST TWO-SIDED WILCOXON p AGAINST ANY SINGLE BASELINE; 7.6×10^{-6} IS THE SMALLEST VALUE THE TEST CAN RETURN AT $n=18$.

Method	BAcc $\uparrow$	ECE $\downarrow$	SCE $\downarrow$	Brier $\downarrow$	NLL $\downarrow$	AURC $\downarrow$	Risk@90 $\downarrow$
Uncalibrated softmax	0.8523 $\pm$ 0.0963	0.0495 $\pm$ 0.0338	0.0468 $\pm$ 0.0311	0.2062 $\pm$ 0.1169	0.3710 $\pm$ 0.1994	0.0486 $\pm$ 0.0510	0.1071 $\pm$ 0.0860
MC dropout [26]	0.8531 $\pm$ 0.0942	0.0571 $\pm$ 0.0337	0.0497 $\pm$ 0.0297	0.2088 $\pm$ 0.1166	0.3773 $\pm$ 0.1987	0.0488 $\pm$ 0.0509	0.1078 $\pm$ 0.0859
Temperature scaling [4]	0.8523 $\pm$ 0.0963	0.0328 $\pm$ 0.0278	0.0393 $\pm$ 0.0289	0.2015 $\pm$ 0.1135	0.3597 $\pm$ 0.1931	0.0484 $\pm$ 0.0510	0.1066 $\pm$ 0.0859
Vector scaling [4]	0.8583 $\pm$ 0.0934	0.0311 $\pm$ 0.0244	0.0313 $\pm$ 0.0252	0.1931 $\pm$ 0.1083	0.3488 $\pm$ 0.1821	0.0444 $\pm$ 0.0472	0.1000 $\pm$ 0.0808
Isotonic regression [6]	0.8569 $\pm$ 0.0941	0.0347 $\pm$ 0.0248	0.0328 $\pm$ 0.0265	0.1968 $\pm$ 0.1113	0.4466 $\pm$ 0.2864	0.0484 $\pm$ 0.0500	0.1018 $\pm$ 0.0829
Dirichlet calibration [5]	0.8580 $\pm$ 0.0960	0.0299 $\pm$ 0.0228	0.0304 $\pm$ 0.0249	0.1930 $\pm$ 0.1089	0.3443 $\pm$ 0.1853	0.0447 $\pm$ 0.0477	0.1014 $\pm$ 0.0829
APEX (ours)	0.8941 $\pm$ 0.0854	0.0339 $\pm$ 0.0206	0.0297 $\pm$ 0.0225	0.1432 $\pm$ 0.0956	0.2589 $\pm$ 0.1635	0.0271 $\pm$ 0.0329	0.0652 $\pm$ 0.0678
APEX better in	17–18/18	6–16/18	9–18/18	18/18	18/18	18/18	17–18/18
worst p vs. any row	$< 10^{-4}$	0.87	0.80	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-3}$

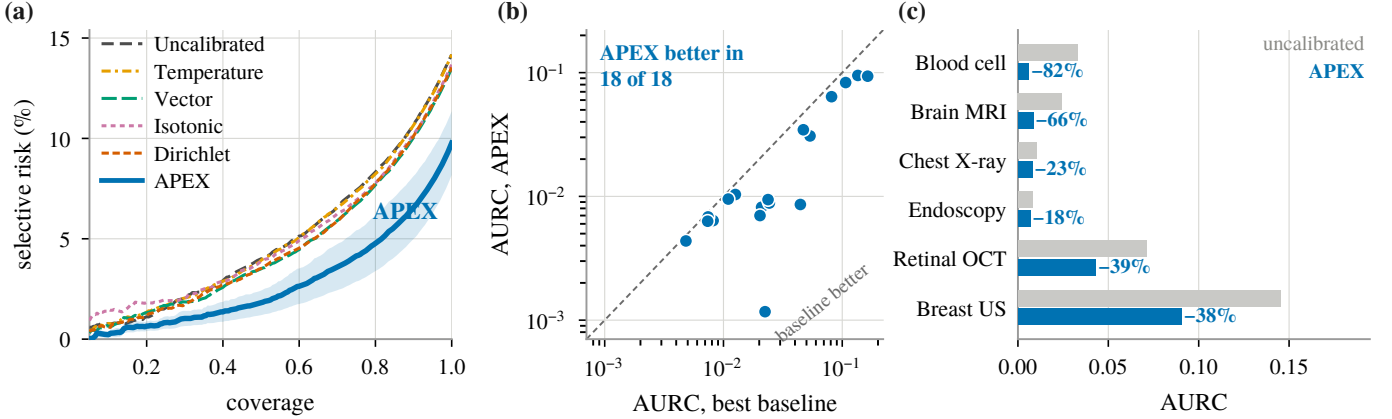


Fig. 2. Selective performance. (a) Mean risk-coverage curve over the 18 configurations, traced from the per-sample predictions of all 90 folds; the band is one standard error. (b) Per-configuration AURC against the best baseline for that configuration. (c) AURC per modality, averaged over the three backbones. Panels (b) and (c) use the same per-fold values as Table I.

error of anything we ran. Fig. 3(a) shows why: APEX sits above the diagonal across the whole range, predicting less confidence than it earns, while Dirichlet tracks it closely. Under-confidence is the direction a triage system can live with, but ECE charges for it either way. What APEX gets right is the global level: its average confidence is within 0.9 points of its accuracy, closer than any other method. The ECE it pays comes from spread inside the bins, not from being systematically too sure of itself.

The divergence has a structural reason that does not depend on our data.

Proposition 1: Let g be strictly increasing on $[0, 1]$. Replacing every confidence c_i by $g(c_i)$ leaves the risk-coverage curve unchanged, and with it AURC, E-AURC and selective risk at every coverage. Expected calibration error has no such invariance.

Proof: At coverage κ the curve reports the empirical error over the $\lceil \kappa n \rceil$ samples of largest confidence, so it depends on the c_i only through the ordering they induce, which a strictly increasing g preserves. ECE compares confidence values with accuracies, $\sum_b (n_b/n) |\bar{c}_b - \bar{a}_b|$, so it moves when those values move. ■

The prediction is exact and it holds numerically: applying c^2 , $\sqrt{c}$, $\sigma(6(c - \frac{1}{2}))$ and $\frac{1}{2} + \frac{1}{2}c$ to the uncalibrated confidences

of all 18 configurations leaves AURC identical in every digit, largest deviation 0, while ECE ranges from 4.95×10^{-2} to 1.44×10^{-1} . A rescaling that cannot change which cases a system defers moves the calibration error it reports by a factor of 2.9.

The consequence shows up in the rankings, and Fig. 3(b) quantifies it. Across the 18 configurations APEX holds mean rank 1.00 on AURC, on Brier and on NLL—it is first in every single configuration on all three—and mean rank 3.78 on ECE, behind Dirichlet, vector scaling and temperature scaling. Choosing a method by calibration error would have selected the wrong method for deferral in 15 of the 18 configurations. A post-hoc layer should be evaluated against the criterion it will be deployed under, and for abstention that criterion is the ordering, not the average gap.

C. What each component contributes

Fig. 4(a) removes one part at a time and re-fits. The RBF expert is load-bearing: without it AURC rises by 2.9×10^{-3} and ECE by 9.0×10^{-3} ($p < 10^{-3}$ for both). The k -NN expert is second for ordering ($+1.5 \times 10^{-3}$ AURC, $p = 2.5 \times 10^{-4}$) and costs a little calibration error to include—a component that helps the ranking while mildly hurting the average gap, the same asymmetry seen at the method level. The cubic SVM and

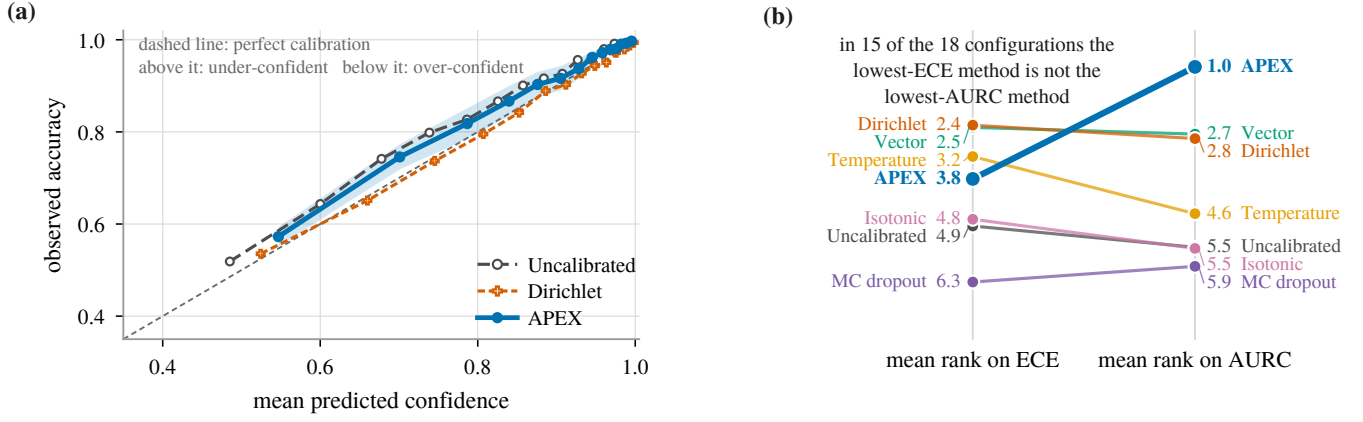


Fig. 3. Calibration and ordering are not the same property. (a) Reliability with equal-mass bins formed inside each configuration and averaged over the 18, the same way the ECE column of Table I is computed; band is one standard error. (b) Mean rank over the 18 configurations on ECE and on AUCR; APEX is first on AUCR in every configuration and mid-field on ECE, and the other methods reorder between the two axes as well.

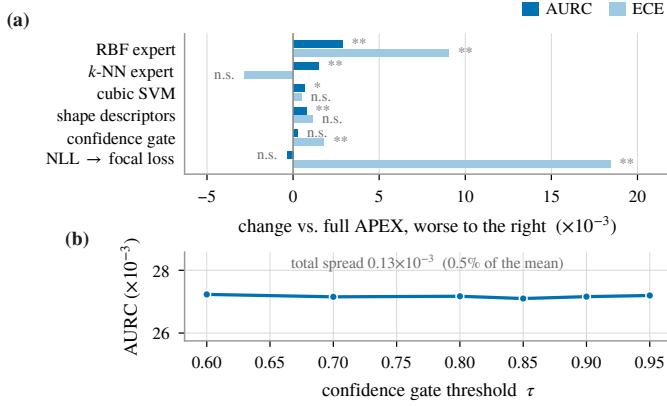


Fig. 4. (a) Change in AUCR and ECE when one component is removed; bars to the right are degradations. Wilcoxon: $**p < 0.01$, $*p < 0.05$. (b) τ is not a tuned parameter—AUCR varies by 0.5% of its own value across the sweep.

the descriptor block each give a small but significant AUCR improvement ($p = 1.6 \times 10^{-2}$ and 1.6×10^{-3}). The gate’s contribution is specific: it improves ECE reliably ($p = 8.4 \times 10^{-4}$) but leaves AUCR unchanged ($p = 0.20$).

No expert reproduces the fused result alone: the RBF SVM, the strongest, reaches 3.12×10^{-2} against 2.71×10^{-2} for the fusion ($p = 1.5 \times 10^{-5}$), and an equal-weight blend without gating or fitted weights reaches 2.85×10^{-2} .

The largest single effect is the objective. Refitting the same experts and gate under a focal loss instead of Eq. 4 raises ECE by 1.8×10^{-2} and NLL by 1.3×10^{-2} ($p = 7.6 \times 10^{-6}$ for both) while leaving AUCR and accuracy statistically unchanged: the fused probabilities are only as honest as the rule used to fit them. The gate threshold is not tuned—sweeping τ over $[0.60, 0.95]$ moves AUCR by 0.5% of its own value (Fig. 4(b)).

D. Full metric detail

Table II reports all fifteen metrics per backbone. Two patterns hold across architectures: the selective and proper scores improve for ViT, ConvNeXt and MobileNetV3 alike,

with the largest absolute gains on the weakest backbone, while maximum calibration error and over-confidence error are the two metrics on which APEX trails even the uncalibrated model on every backbone.

VI. DISCUSSION

The reason is simple. A frozen linear head is a projection, and projections lose information: two samples with nearly the same logits can sit in very different neighbourhoods of the latent space—one deep inside a class cluster, the other on a boundary—and the softmax cannot express the difference. The kernel experts can, an RBF machine reading local density and a k -NN classifier neighbourhood purity, so fusing their posteriors with the softmax reinstates a signal about *which* predictions are unsafe.

It also explains the asymmetry. A parametric calibrator does the opposite of what APEX does—temperature scaling provably cannot change any prediction, so it moves ECE while leaving the ordering untouched. Proposition 1 is the formal statement of that separation, and it is why the two families are complementary: composing a scaling layer on a fusion layer is a natural next step we have not tested.

The numbers translate directly: at 90% coverage selective risk falls from 10.71% to 6.52%, so roughly two in five of the errors that would have reached a clinician are routed for review instead, at no change to the network.

VII. LIMITATIONS

Four limitations bound these claims. The backbones are frozen and only linear heads trained; we have not shown the gains survive full fine-tuning, where a stronger head may already encode some of the geometry the experts recover. That is the most important experiment we did not run.

Every result is from a single seed—fold assignment, the fit/calibration split and head initialisation are fixed—so the 18-of-18 margins carry unmeasured seed variance. The breast ultrasound corpus is small (778 images after the audit) and

TABLE II

ALL FIFTEEN METRICS PER BACKBONE, AVERAGED OVER THE SIX DATASETS (EACH ENTRY IS ITSELF A MEAN OVER FIVE FOLDS). *None* IS THE UNCALIBRATED SOFTMAX; DIRICHLET IS THE STRONGEST BASELINE BY MEAN RANK ON ECE, BRIER AND NLL. ARROWS GIVE THE PREFERRED DIRECTION; BEST OF EACH BLOCK IN BOLD.

Metric	Dir.	ViT-S/16 ($d=384$)			ConvNeXt-T ($d=768$)			MobileNetV3-L ($d=1280$)		
		None	Dirich.	APEX	None	Dirich.	APEX	None	Dirich.	APEX
Balanced accuracy	↑	0.8620	0.8653	0.9040	0.8621	0.8777	0.8961	0.8329	0.8310	0.8822
Macro F1	↑	0.8630	0.8679	0.9073	0.8632	0.8751	0.8981	0.8280	0.8311	0.8839
Macro AUROC	↑	0.9606	0.9617	0.9769	0.9656	0.9659	0.9769	0.9496	0.9489	0.9692
ECE	↓	0.0394	0.0297	0.0333	0.0455	0.0299	0.0329	0.0635	0.0301	0.0355
Debiased ECE	↓	0.0150	0.0072	0.0127	0.0212	0.0070	0.0126	0.0362	0.0078	0.0143
Class-wise SCE	↓	0.0409	0.0290	0.0279	0.0440	0.0293	0.0301	0.0555	0.0329	0.0312
Maximum CE	↓	0.3329	0.3588	0.3769	0.3738	0.2894	0.3808	0.3030	0.3244	0.3464
Over-confidence error	↓	0.6646	0.7071	0.6842	0.6561	0.7083	0.6874	0.6056	0.6749	0.6783
Brier score	↓	0.1938	0.1833	0.1269	0.1858	0.1734	0.1385	0.2389	0.2224	0.1641
Brier skill score	↑	0.7208	0.7356	0.8142	0.7323	0.7497	0.7987	0.6561	0.6799	0.7619
NLL	↓	0.3473	0.3290	0.2311	0.3327	0.3081	0.2495	0.4330	0.3957	0.2962
AURC	↓	0.0463	0.0414	0.0244	0.0390	0.0360	0.0243	0.0606	0.0566	0.0326
Excess AURC	↓	0.0333	0.0295	0.0178	0.0265	0.0252	0.0168	0.0404	0.0376	0.0233
Risk at 90% coverage	↓	0.0993	0.0923	0.0558	0.0938	0.0872	0.0614	0.1284	0.1248	0.0784
Risk at 80% coverage	↓	0.0748	0.0688	0.0396	0.0699	0.0653	0.0433	0.1025	0.0971	0.0586

drives the widest fold-to-fold spread; all six corpora are public, with no external or prospective validation.

Finally, APEX does not improve expected calibration error over the best parametric baselines and is worse than one of them. A deployment needing a calibrated probability should use a calibrator, or compose one with APEX; our claim is confined to the ordering of confidence and the abstention decisions that depend on it.

VIII. CONCLUSION

We asked what a post-hoc layer can do for a frozen medical image classifier built for abstention rather than average calibration. Reading the latent representation the classifier discards, fitting three kernel experts over a shared projection of it and mixing their posteriors with the softmax under a confidence-conditional gate reduces AURC by 44.3% and wins all 18 configurations against each of six baselines, with Brier and log-likelihood doing the same and balanced accuracy winning at least 17 of 18. Expected calibration error does not improve beyond what parametric calibrators achieve—the consequence of optimising a different property, not a shortfall to explain away. Selective risk is invariant to monotone rescaling where ECE is not, and the two disagree about which method to deploy in 15 of 18 cases; the criterion should be chosen before the method is.

REFERENCES

- [1] B. Kompa, J. Snoek, and A. L. Beam, “Second opinion needed: communicating uncertainty in medical machine learning,” *npj Digit. Med.*, vol. 4, no. 1, p. 4, 2021.
- [2] R. El-Yaniv and Y. Wiener, “On the foundations of noise-free selective classification,” *J. Mach. Learn. Res.*, vol. 11, pp. 1605–1641, 2010.
- [3] Y. Geifman and R. El-Yaniv, “Selective classification for deep neural networks,” in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.
- [4] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. ICML*, 2017, pp. 1321–1330.
- [5] M. Kull *et al.*, “Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with Dirichlet calibration,” in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
- [6] B. Zadrozny and C. Elkan, “Transforming classifier scores into accurate multiclass probability estimates,” in *Proc. ACM SIGKDD*, 2002, pp. 694–699.
- [7] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, “Obtaining well calibrated probabilities using Bayesian binning,” in *Proc. AAAI*, vol. 29, 2015, pp. 2901–2907.
- [8] A. Kumar, P. S. Liang, and T. Ma, “Verified uncertainty calibration,” in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
- [9] J. Nixon, M. W. Dusenberry, L. Zhang, G. Jerfel, and D. Tran, “Measuring calibration in deep learning,” in *Proc. CVPR Workshops*, 2019, pp. 38–41.
- [10] G. W. Brier, “Verification of forecasts expressed in terms of probability,” *Mon. Weather Rev.*, vol. 78, no. 1, pp. 1–3, 1950.
- [11] T. Gneiting and A. E. Raftery, “Strictly proper scoring rules, prediction, and estimation,” *J. Am. Statist. Assoc.*, vol. 102, no. 477, pp. 359–378, 2007.
- [12] Y. Geifman and R. El-Yaniv, “SelectiveNet: A deep neural network with an integrated reject option,” in *Proc. ICML*, 2019, pp. 2151–2159.
- [13] D. Hendrycks and K. Gimpel, “A baseline for detecting misclassified and out-of-distribution examples in neural networks,” in *Proc. ICLR*, 2017.
- [14] D. H. Wolpert, “Stacked generalization,” *Neural Netw.*, vol. 5, no. 2, pp. 241–259, 1992.
- [15] J. Platt, “Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods,” in *Advances in Large Margin Classifiers*. MIT Press, 1999, pp. 61–74.
- [16] J. Mukhoti *et al.*, “Calibrating deep neural networks using focal loss,” in *Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 15 288–15 299.
- [17] P. Mooney, “Blood cell images,” Kaggle, kaggle.com/datasets/paultimothymooney/blood-cells (accessed Aug. 2026).
- [18] M. Nickparvar, “Brain tumor MRI dataset,” Kaggle, kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset (accessed Aug. 2026).
- [19] D. S. Kermany, M. Goldbaum, W. Cai *et al.*, “Identifying medical diagnoses and treatable diseases by image-based deep learning,” *Cell*, vol. 172, no. 5, pp. 1122–1131, 2018.
- [20] Heartzhacker, “Medical imaging,” Kaggle, kaggle.com/datasets/heartzhacker/medical-imaging (accessed Aug. 2026).
- [21] W. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, “Dataset of breast ultrasound images,” *Data in Brief*, vol. 28, p. 104863, 2020.
- [22] B. Barz and J. Denzler, “Do we train on test data? Purging CIFAR of near-duplicates,” *J. Imaging*, vol. 6, no. 6, p. 41, 2020.
- [23] A. Dosovitskiy *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.
- [24] Z. Liu *et al.*, “A ConvNet for the 2020s,” in *Proc. CVPR*, 2022, pp. 11 976–11 986.
- [25] A. Howard *et al.*, “Searching for MobileNetV3,” in *Proc. ICCV*, 2019, pp. 1314–1324.
- [26] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian approximation: Representing model uncertainty in deep learning,” in *Proc. ICML*, 2016, pp. 1050–1059.
- [27] J. Demšar, “Statistical comparisons of classifiers over multiple data sets,” *J. Mach. Learn. Res.*, vol. 7, pp. 1–30, 2006.
- [28] F. Wilcoxon, “Individual comparisons by ranking methods,” *Biometrics Bull.*, vol. 1, no. 6, pp. 80–83, 1945.